



LENS

Learning Engineering for Next-Generation Systems

The Ten Courses, In Depth

Curriculum development document for the LDT concentration

William Gray-Roncal, PhD

September 11, 2026 · Working draft for program-director review

Not a commitment of record. The basis for the decisions listed in the consolidation section.

Contents

Overview The three commitments and the decisions required.

Part 1. The Frame Audience, differentiation from LXD and AILE, the layered recruiting message, the objective legend, and the two structural moves.

Part 2. The Core: LEN 1, 2, 3 The three required courses, rebalanced as a set. Measurement as the spine; governance and bias folded in.

Part 3. Methods and Capstone: LEN 4, 6, 10 The shared and built methods courses and the capstone. The iteration ladder.

Part 4. The Four Electives: LEN 5, 7, 8, 9 Depth courses; what each folded forward; the two-slot logic.

Part 5. Consolidation and Decisions The cut-lines, record changes, elective structure, decisions, advising, and build sequence.

Overview

The concentration is ten courses: two inherited (F1, F2), one shared methods course (LEN 4), three core (LEN 1–3), one LENS-built methods course (LEN 6), four electives (LEN 5, 7, 8, 9), and a capstone (LEN 10). This pass rebalanced them around three commitments.

- **One identity, enterable by the actual pool.** LENS recruits people who want to think in systems and evidence, not engineers. The register is a stance, not a domain; high-consequence work is the ceiling, not the entry bar. Measurement is the spine, because it carries the distinction from LXD and is where the jobs are.
- **Essentials live in required courses, never in skipped electives.** Governance and bias fold from LEN 7 into LEN 2 and LEN 3; implementation essentials route from LEN 8 into the LEN 10 capstone as lectures. Electives then carry depth.
- **LEN 1 becomes the gateway.** Reduced from seven objectives to five (frame, measure, build), it sheds advanced method to LEN 3 and LEN 5 and becomes enterable by all three concentrations.

Six decisions are required, listed in Section 4. The two that gate the rest: whether LEN 1 launches on the redesigned syllabus, and whether LEN 5 is required rather than recommended. Record changes touch doc 1 and doc 2 and are listed in Section 2.

The Frame

1. The two facts on the ground

Two facts constrain every decision in this pass, and the earlier drafts of the field map ignored both. First, the concentration lives in a School of Education with zero engineers, and its applicant pool is educators, instructional designers, and analysts, not engineers. A concentration positioned as “for engineers moving into learning” is fishing in an empty pond inside its own building. Second, relevance is set by the job market, not by a standards taxonomy. Whatever competencies LENS names have to map to roles a graduate can actually be hired into. These two facts together rule out the version of LENS that reads as an engineering program transplanted into a school of education. What survives is a concentration defined by a way of thinking, not by a prior credential or a domain.

2. Who LENS is for: a disposition, not a background

LENS is for people who want to think in systems and evidence. Not “engineers.” The student is someone, from any background in the pool, who is not satisfied asking whether a learning experience is engaging, and wants to ask how the whole system produces the outcome and how they would know. That disposition already exists among educators, designers, and analysts who feel underserved by a pure design track or a pure leadership track. It is recruitable inside a school of education because it is a way of thinking, not a degree someone arrives with. Engineering, in this framing, is a stance, rigor, measurement, systems thinking, not a credential and not a domain. “Capability at the operator-system interface” is the advanced form of the disposition, reached by the end of the concentration, not the price of entry.

All three concentrations sit in the LDT M.Ed., share the two inherited courses and the shared methods course, and carry the same six program objectives. They differ less in who enters them than in the altitude and rigor of the question each trains you to ask.

	LXD	AILE	LENS
Trains you to ask	Does this experience work for these learners?	How should this institution adopt and lead AI in learning?	How does the whole system produce the outcome, and how would I know?
Center of gravity	Experience craft	Institutional leadership	Systems thinking and analytics
Unit of analysis	The learner and the experience	The organization and its AI strategy	The system and its measurable outcomes
Stance	Human-centered design	Leadership and governance	Measurement and systems rigor (an engineering stance, not a domain)
Signature output	A designed experience	An adoption or governance plan	A measured, iterated intervention with evidence
Hires as	Learning experience designer, ID	Ed-tech lead, L&D director, AI strategy	Learning engineer, learning analytics, measurement and evaluation, L&D strategy

The differentiator, in one line. LXD trains the craft of the experience. AILE trains the leadership of adoption. LENS trains systems thinking and analytics: seeing the whole system and insisting on evidence. The distinction rests on altitude and rigor, not on domain, which is what lets it recruit from the pool that actually exists. **High-consequence work does not leave; it changes role.** It stops being the entry requirement, which no one in the pool can meet, and becomes the ceiling: the place where the systems-and-evidence rigor a student is learning gets tested at its hardest, where a capability gap is paid by someone who did not choose the risk. An educator can enter LENS without ever touching a cockpit and still be drawn by the fact that the same methods hold where lives are on the line. That is what makes the rigor feel real rather than academic. So the distinction now rests on two things together: the analytics-and-measurement axis, which is where the jobs are, and high-consequence as the proof that the rigor matters. Measurement is the spine; high-consequence is the ceiling.

3. Boundaries are gradients, and that protects participation

The map draws clean lines to be legible, not because the concentrations are sealed. A student can sit legitimately in two framings; a course can be genuinely co-owned; a designer, a leader, and an analyst can be on the same problem and all be doing real work. Naming the center of each concentration protects it; it does not police the edges. So this pass does not treat “this could sit in LXD or AILE” as a reason to cut or to close a course off. Much of the best and most enrollable work sits exactly on the overlap. The cut test is narrower and is stated on every card: does removing this leave LENS unable to teach systems thinking and evidence at the altitude that defines it? If yes, it stays regardless of overlap. If no, overlap is a feature, and often a recruiting asset.

4. Where the three genuinely converge

Synergy is real, and naming it precisely is what lets LENS open some courses outward without dissolving. Three convergence zones:

Zone	What is shared	Who leads it
Evidence and measurement	Every concentration needs students who can read and produce decision-grade evidence. This is the shared methods course's job, and LENS injects the capability-measurement view into it.	Shared methods course; LENS contributes
Human-machine collaboration	AI as a partner in learning is AILE's leadership question and LENS's design question. The same phenomenon, approached as govern-it versus design-it. This is the richest synergy, and LEN 2 is where it lives.	LENS (design) and AILE (leadership), naturally paired
The learning environment as a system	An LXD experience is a system with operators and an impact; the capability frame applies to it directly. LENS gives LXD a systems-and-measurement lens on what they already design.	LENS offers the lens; LXD owns the experience

The synergy rule the cards apply. A LENS course is offerable outward when it teaches one of these shared things at an altitude the sibling concentrations do not reach, and stays LENS-centered when it teaches the concentration's own methods, systems modeling, measurement design, iteration against evidence. Convergence is where LENS contributes to the program and recruits from it; distinction is where LENS earns its name.

LENS is AI-practitioner-aware, and does not cede AI to AILE

AI is not AILE's private territory. AILE owns the leadership and governance of AI adoption. LENS owns something adjacent and equally market-relevant: building and measuring AI-augmented capability interventions, and evaluating whether they work. The field's own community has moved this way, learning engineering is now being applied to AI system development, not only AI applied to learning. A LENS graduate should be able to instrument an AI tutor, judge an LLM-as-evaluator design, and say honestly whether an AI intervention built capability or quietly eroded judgment. That is an AI-practitioner competency, it is where jobs are growing, and it is LENS's to claim as the design-and-measure counterpart to AILE's lead-and-govern. It lives in LEN 2 (human-machine teaming) and LEN 9 (computational and AI methods), and it is named on those cards.

5. Recruiting message: one identity, layered over years

The audience is both, and the message is layered rather than split. LENS has to recruit the systems-and-analytics-minded people already in the school, and, over time, the engineers and career-changers who make it distinctive. Those are not two programs and not two messages held at equal volume. They are one identity, systems thinking and evidence, with an inclusive door and a distinctive ceiling, and a mix that shifts deliberately across a multi-year runway.

The layered message

Layer	What it says
The door (inclusive)	For educators, designers, and analysts who want to think in systems and evidence, not just craft experiences.

Layer	What it says
	You do not need to be an engineer. You need to want to ask how the whole system produces the outcome, and how you would know.
The room (the work)	You learn to see a learning environment or an operational system as a system, measure whether it builds the capability it promises, and iterate against evidence. You leave able to instrument, analyze, and defend an intervention.
The ceiling (distinctive)	These methods hold where the stakes are highest, in health, aviation, defense, and education at scale, where a capability gap is paid by people who did not choose the risk. That is what makes the rigor real, and it is where the field is heading.

The discipline that keeps “both” from becoming “nothing.” The audience widens; the disposition does not. LENS recruits inclusively for people who want to think in systems and evidence, and it does not dilute what that means to accommodate people who do not. Inclusive on who may enter; firm on what they will become.

The multi-year arc

Phase	Emphasis	Why
Years 1–2	Lead with the door. Recruit the systems-and-analytics-minded from the SoE pool. High-consequence present as the ceiling, not the gate.	This is who can actually be enrolled now, in a building with no engineers. Fills the first cohorts and clears the review threshold.
Years 2–4	Grow the distinctive emphasis. As graduates, the convening, and the APL practicum become proof points, message toward engineers, career-changers, and technical practitioners.	The distinctive audience needs evidence the program is real before they take the risk. Proof points earn the right to that message.
Throughout	High-consequence stays present the whole time, as ceiling and credibility, never dropped and never the entry bar.	It is what makes LENS unmistakably not-LXD, and dropping it would forfeit the distinction the analytics axis cannot carry alone.

This section is a summary. A fuller recruiting-strategy document, with channels, target lists, the standing call for students (blocker B2 in the goals), and the primer for cross-concentration entry, follows separately. What matters for the course cards is the identity: every course is built for the systems-and-evidence disposition, enterable without an engineering background, with high-consequence as the ceiling it reaches toward.

6. The objective legend

Every course card in Parts 2–5 states its PLO and LEO coverage explicitly, at subobjective grain, and marks the depth. This legend defines the terms so the cards can be terse.

The two objective systems

PLOs (Program Learning Objectives) are the LDT program’s six, shared by all three concentrations, filed with MHEC. Every LENS course must map to them, because that is the accreditation frame. They are broad: theory and evidence in design (1), ethical and human-centered solutions (2), collaboration and leadership (3), communication (4), sociocultural aspects (5), data-informed design (6).

LEOs (LENS Educational Objectives) are the concentration’s own five, held in doc 1: Systems Analysis (1), Iterative Development (2), Human-System Collaboration (3), Data, Measurement, and Evaluation (4), Navigating Sociotechnical Constraints (5). They are the specialization’s spine and are what the attainment profile aggregates. PLOs satisfy the program; LEOs define the concentration.

The LEOs are triangulated, not invented. They draw on three intellectual traditions that converge on the same object: the learning sciences (how people develop), the IEEE ICICLE learning-engineering framework (the field’s own competency and process guidance), and the INCOSE capability-engineering subdomain of systems engineering (engineering a capability across a lifecycle and a system-of-systems, rather than engineering a single product). LENS sits where those three meet: it is, in effect, human-capability engineering, the systems-engineering capability subdomain with the person moved from the margin to the center. This is a touchpoint that grounds the objectives and shapes LEN 3; it is deliberately kept out of the front-facing recruiting message, where the engineering register would narrow the audience the concentration needs.

Depth marks

Mark	Meaning
Assess	The course assesses this subobjective at practitioner depth; a graded artifact carries it. This is where the attainment profile draws evidence.
Introduce	The course opens the subobjective at awareness; a later course assesses it. Named so the profile does not over-claim.
Reinforce	The subobjective was assessed earlier; this course exercises it again without being its assessment home.

A course that assesses a LEO subobjective must also show which PLO subobjective that work satisfies, because the syllabus speaks in CLO and PLO while the record holds the CLO-to-LEO join. The cards show both.

7. The card format used in Parts 2–5

Each course gets one card with eight rows, in this order:

- **Owns.** The one thing this course exists to teach that no other course does.
- **Major topics.** The draft module-level topics, as a sequence.
- **Assignments.** The draft graded and ungraded work.
- **PLO coverage.** Which PLO subobjectives, at which depth.
- **LEO coverage.** Which LEO subobjectives, at which depth.
- **AILE / LXD relationship.** Distinct from, converges with, or offerable to, per the field map.
- **If we must cut.** The honest line: what leaves first, and what it costs. This is the row that makes the document useful under pressure.
- **Flags.** Dependencies and decisions.

8. Two structural moves the cards assume

Both come from your direction and both reshape multiple cards, so they are stated once here rather than re-argued on each card.

Move A: LEN 7 is not load-bearing, so its essentials fold forward

The problem. Most students take two electives, and if LEN 5 and LEN 9 are the recommended pair, most never take LEN 7. Governance, bias, and risk cannot live only in a course most students skip, or most LENS graduates will never study the ethics of the systems they build. That is indefensible for a concentration whose own materials treat ethics as a design constraint.

The move. Pull the essentials of LEN 7 into the required core. Draft split, to confirm on the LEN 2 and LEN 3 cards: governance and accountability (who is responsible, what oversight exists, when good design should say do not deploy) into LEN 2, where human-machine authority already lives; bias, fairness, and measurement equity into LEN 3, where systems and measurement of the interface already live. LEN 7 remains as the depth elective for students who want the full treatment, but it stops being the only home for the essentials.

The cost, named. LEN 2 and LEN 3 get fuller, which collides with the fact that LEN 1 is also still too full. The core sequence has to be rebalanced as a set, not course by course. Parts 2 handles all three core cards together for exactly this reason.

Move B: LEN 8 feeds LEN 10, and LEN 10 can lecture

The move. LEN 10, the capstone, is not overfull and does not offload. It receives: implementation-science content from LEN 8 (adoption, sustainment, why evidence does not move itself) arrives in the capstone as lectures alongside the project, so that a student who never took LEN 8 as an elective still meets its load-bearing ideas at the moment they matter, standing up a real intervention at a partner. This makes LEN 8, like LEN 7, safe to be a true elective: its essentials reach everyone through LEN 10.

The pattern under both moves. Anything essential cannot live only in an elective. It either folds into the core (LEN 7 into 2 and 3) or arrives through a required course as received content (LEN 8 into 10). Electives then carry depth, not essentials. That principle is the backbone of the whole pass, and the cut-lines in Parts 2–5 enforce it.

9. Next

Part 2 takes the three core courses (LEN 1, 2, 3) together, in full depth, because Move A rebalances them as a set: LEN 1 sheds to get un-full, LEN 2 gains human-machine governance, LEN 3 gains bias and measurement equity. Each gets the full eight-row card. Parts 3 (LEN 4, 6, 10), 4 (electives 5, 7, 8, 9), and 5 (the consolidated cut-line table and the record changes) follow.

One decision to carry into Part 2: the Move A split (governance to LEN 2, bias to LEN 3). If you would rather both go to LEN 3, or split differently, say so and Part 2 builds to that. Everything else in Part 1 is a frame you can accept as-is or adjust.

PART 2

The Core: LEN 1, 2, 3

The core rebalance, in one view

Read these three cards as a system. The same two forces move through all of them.

Force	LEN 1	LEN 2	LEN 3
Load change	Sheds (failure methods, lifecycle, governance diagnostic already left in the redesign)	Gains governance and accountability (Move A)	Gains bias, fairness, measurement equity (Move A)
Measurement role	Introduces "what counts as evidence" (new CLO 4)	Measures the collaboration itself	Measures the interface; fairness as a measurement property
Net fullness	Right, after shedding	Full: watch it	Full: watch it

The tension to hold across the three: Move A makes LEN 2 and LEN 3 fuller at the same time the redesign is trying to make the sequence lighter. The resolution is that what LEN 1 sheds is not the same kind of thing LEN 2 and LEN 3 gain: LEN 1 sheds advanced method (failure taxonomies, lifecycle models) that was premature in a first course; LEN 2 and LEN 3 gain governance and fairness essentials that every graduate must meet. The sequence gets lighter at the front and denser in the middle, which is the right shape. But LEN 2 and LEN 3 are now at their ceiling; nothing else may be added to them without a cut, and each card says what its cut is.

LEN 1: Principles of Learning Engineering for Complex Systems

LEN 1 Core 1 15 weeks · 6 modules · 5 CLOs · LEO-1, 2; LEO-4 introduced

Owns	The gateway. Three things and only three: frame (capability as the interface between what a system requires of its people and the impact it delivers), measure (what would count as evidence of a capability, before building), and build (one iteration cycle on the student's own system). It is the course that turns a systems-and-analytics disposition into a method. It owns the definition the whole concentration turns on.
Major topics	● M1 Framing Capability (Wks 1–2). Capability as interface; the reframe from technology/training problem; two narrated cases. ● M2 What the Science of Learning Contributes (Wks 3–4). Cognitive load and motivation, recast around stakes; the sociocultural lens. ● M3 Seeing the System (Wks 5–6). Components, interfaces, feedback loops; where capability decisions occur. Sketch grain, not modeling method. ● M4 Measuring Capability (Wks 7–9). What counts as evidence; operationalizing a construct; designing the assessment before the content; the flywheel. The spine module. ● M5 The Iteration Cycle (Wks 10–13). Design, instrument, evaluate, refine, on the student's own system. AI-augmented interventions welcome. ● M6 Making the Case (Wks 14–15). Communicating an iteration and its evidence to a mixed audience; one coupling failure as anchor.
Assignments	● Domain Expertise Inventory (ungraded gate, Wk 1): the system carried all term. ● Module 1 Assignment: How Learning Engineering Succeeds or Fails (graded, Wk 2): one success and one failure case, plus the conditions synthesis. ● Assignment 1: Capability Measurement Plan (Wk 9): what would count as evidence, operationalized, before building. ● Assignment 2: Iteration Cycle Report (Wk 13): one complete documented iteration on the student's own system. ● Final: Iteration Briefing (Wk 15): the iteration and its evidence, to a mixed audience. ● Weekly discussions and labs (participation): the reflection lives here.
PLO coverage	Assess: 1.3, 1.4 (theory-to-design); 2.2 (systems thinking for ethical tools); 6.1, 6.4 (data-informed design); 4.2 (communication, via the briefing).

	Introduce: 1.5 (deploy grounded solutions); 3.2 (communicate shared visions); 6.2 (revise on data).
LEO coverage	Assess: 1.1 (capability requirements); 2.1, 2.2, 2.5 (iteration). Introduce: 1.2, 1.3 (system modeling, at sketch grain); 4.8, 4.1, 4.5 (measurement design, opened here, assessed in the shared methods course). The measurement-as-spine consequence: LEN 1 now carries LEO-4 at introduce depth, which it did not before. This is the first course to put data-informed design in front of every student, and it is what makes the analytics axis load-bearing from week one.
AILE / LXD	Open, and the primary shared entry. The frame applies to a learning environment (LXD) and an AI-augmented intervention (AILE) without translation. Recommended as the door for cross-concentration students; a self-paced primer version exists for those entering LEN 2 or LEN 7 without it. This is where the inclusive message lands.
If we must cut	M3 (Seeing the System) compresses to one week before anything else goes, since LEN 3 models the system for real. Below that, M2's sociocultural segment could move to a reading rather than a taught session. What must not be cut: M4 (measurement), because it is the spine, and the iteration cycle in M5, because it is the build. Cutting either would turn LEN 1 back into a survey.
Flags	Running-this-term decision still open (After vs Before syllabus). M4 needs its own peer-reviewed measurement sources, which replaced the failure-analysis anchors and must be found, not assumed. The redesign is built; this card is the redesigned course.

LEN 2: Human-Machine Teaming and Adaptive Learning Systems

LEN 2 Core 2 *LEO-3 (all) + governance, folded from LEN 7 · builds on LEN 1*

Owns	The collaboration as the unit of design and measurement. Where the human-machine team, not the operator alone, is the thing engineered: role architectures, authority, delegation with revocation, interface and alert transparency, AI as creative partner and epistemic risk, automation bias and skill atrophy. After Move A, it also owns the governance and accountability of that collaboration. This is the richest AILE-convergence course and the clearest place LENS is AI-practitioner-aware.
Major topics	● The collaboration as unit of analysis; role architectures and authority gradients. ● Delegation with revocation; disconfirming evidence named in advance; interface, alert, and mode transparency. ● AI as creative partner and epistemic risk; automation bias, cognitive offloading, skill atrophy. ● Measuring the collaboration: what evidence shows the team is better than either party alone. ● Governance and accountability (folded from LEN 7): who is responsible when a human-machine team fails; what oversight exists; when good design should say do not deploy. The governance-objection diagnostic lands here as a required topic, not an elective one. ● Facilitators as roles (3.7).
Assignments	● A collaboration design for an adaptive or AI-augmented system, with the authority gradient and revocation conditions specified. ● A measurement plan for the collaboration, naming automation-bias and offloading risks and how they would be detected. ● A governance memo (from the folded LEN 7 material): for one design decision, distinguish a governance objection good design can dissolve from one that correctly says do not deploy.
PLO coverage	Assess: 3.1, 3.3 (collaboration and quality leadership); 2.2 (systems thinking, ethical tools); 5.2 (power dynamics, via governance). Introduce/Reinforce: 6.3 (mixed-methods analysis of the collaboration); 2.3 (equity of the collaboration, reinforced).
LEO coverage	Assess: all of LEO-3 (3.1–3.7): human-system collaboration is this course's domain. Introduce: 5.x governance subobjectives, via the folded material (governance as a sociotechnical constraint).
AILE / LXD	Converges hardest with AILE, and by design. AILE leads the governance of AI adoption; LEN 2 designs and measures the human-machine collaboration. Same phenomenon, lead-it versus design-it. Strongest candidate to cross-list or co-teach with AILE. For LXD, the interface and alert work is experience design at the boundary. The AI-practitioner competency lives here.
If we must cut	The facilitators-as-roles topic (3.7) is the lightest and goes first. The folded governance material must NOT be cut,

	because that is the whole point of Move A: if governance leaves LEN 2, it returns to being elective-only and most students never meet it. If load forces a choice, cut a teaming subtopic before cutting governance.
Flags	Full after Move A. The bridge problem: an AILE student entering LEN 2 without LEN 1 lacks the frame and the measurement plan LEN 2 assumes; the primer answers this. Confirm the governance-into-LEN-2 half of Move A (the other half, bias, goes to LEN 3).

LEN 3: Learning Engineering Systems

LEN 3 Core 3 *LEO-1, 5 + bias/fairness, folded from LEN 7 · where LENS becomes itself*

Owns	The systems course, and where the specialization gets serious. Modeling capability as interacting systems; integration at speed and scale; the governance seam as a produced artifact. It is where LEN 1's sketch-level "seeing the system" becomes real modeling. After Move A, it also owns bias, fairness, and measurement equity: the fairness properties of the interface and the data that measures it.
Intellectual anchor	Capability-based systems engineering (INCOSE). This is the course where the INCOSE touchpoint lands substantively. Capability engineering, defining a capability as the ability to achieve an effect under stated conditions and engineering it across a lifecycle and a system-of-systems, is exactly LEN 3's modeling discipline, with the human capability moved to the center. Naming the anchor here does three things: it gives LEN 3 an external, credentialed methodology rather than a bespoke one; it connects LENS to the systems-engineering community (INCOSE) as ICICLE connects it to the learning-engineering community; and it is the substance behind the claim that LEN 3 is where LENS becomes unmistakably itself. HSI (Boy & Kennedy) sits alongside it as the human-integration half.
Major topics	● Capability-based SE and HSI frameworks; the vocabulary and stance, applied, not surveyed. ● Modeling capability as interacting systems: components, interfaces, feedback, the lifecycle, the system-of-systems view. ● Rotating domain studios (the practitioner rotation inherited from LEN 1): integration failures worked in context. ● The governance seam as a produced artifact (the xAPI/standards week; takes 5.6). ● Bias, fairness, and measurement equity (folded from LEN 7): algorithmic bias in learning systems; fairness as a measurable property of the interface and its data; fairness beyond omission. Every graduate meets this here, not only the LEN 7 electives. ● First contact with failure-analysis methods, coded against the casebook taxonomy (depth is LEN 5).
Assignments	● A system model of the student's capability problem as interacting systems, with the governance seam specified as a deliverable. ● A domain-studio integration analysis: a real integration failure, modeled and coded. ● A fairness analysis (from the folded material): for one interface, where bias could enter through the system or its measurement, and what a fair-measurement design would require.
PLO coverage	Assess: 2.2 (systems thinking); 5.1, 5.2, 5.3 (sociocultural: bias, power, mitigation, via the folded material); 6.3 (mixed-methods). Reinforce: 1.3, 1.4 (theory-to-design, from LEN 1); 3.4 (evaluation leadership).
LEO coverage	Assess: 1.2, 1.3, 1.4 (systems modeling at practitioner depth); 5.x (navigating sociotechnical constraints, including the folded bias/fairness subobjectives). Reinforce: 1.1 (capability requirements, from LEN 1); 5.6 (the standards/seam artifact). The capability-engineering consequence: LEO-1 (Systems Analysis) is defined by the INCOSE capability-engineering subdomain, and LEN 3 is its practitioner home. This is where the attainment profile draws its LEO-1 evidence at depth.
AILE / LXD	LENS-centered: the systems modeling and domain studios are the concentration's own methods, and this is the course that certifies the claim. But the folded fairness material converges with F2 (critical perspectives) and with AILE's governance concerns; state the division, F2 critiques, LEN 3 measures and designs. Not offered outward as a whole, but the fairness-measurement topic is a natural guest-lecture bridge to AILE.
If we must cut	First contact with failure methods compresses to a single coded example (depth is LEN 5 regardless). Below that, one domain studio rotation drops from three to two. What must not be cut: the folded bias/fairness material, for the same reason as LEN 2's governance, if it leaves LEN 3 it becomes elective-only and most graduates never study fairness; and the system-modeling core, which is the course's reason to exist and LEO-1's assessment home.
Flags	Full after Move A, and the course that absorbed the most from the LEN 1 redesign (lifecycle, HSI, practitioner

	rotation) plus the bias fold-in. Watch total load closely; this is the likeliest course to overflow, and the LEN 5 split (failure methods to LEN 5 at depth) is what keeps it at six modules. Confirm the bias-into-LEN-3 half of Move A. The systems-thinking grain rule (LEN 1 sees, LEN 3 models, LEN 6 frames) must be written into M1 of this course.
--	--

What Part 2 commits, and what Part 3 does

- **Move A is now concrete:** governance and accountability into LEN 2; bias, fairness, and measurement equity into LEN 3. Both marked “must not cut” on their cards, because that is the point, essentials cannot live only in a skipped elective.
- **Measurement is the spine:** LEN 1 introduces LEO-4 in week one; the analytics axis runs through all three core cards, not one.
- **The INCOSE touchpoint landed where it belongs:** in LEN 3’s methodology and in the LEO legend, not in the recruiting message.
- **The core is denser in the middle by design:** LEN 1 sheds advanced method; LEN 2 and LEN 3 gain governance and fairness essentials. Both are now at their ceiling, and each card names its cut.

Part 3 takes LEN 4 (the shared methods course LENS injects into, not ours to redesign), LEN 6 (the LENS-built methods course), and LEN 10 (the capstone that receives from LEN 8 and can lecture). Part 4 takes the four electives. Part 5 consolidates the cut-lines and the record changes.

PART 3

Methods and Capstone: LEN 4, 6, 10

One idea that spans these three: the iteration ladder

The redesign created a ladder nobody planned, and Part 3 is where it becomes visible. The same iteration cycle is run three times, at escalating authenticity. Naming it is what keeps the three from feeling redundant.

	LEN 1	LEN 6	LEN 10
Iterates on	The student's own system	An authentic cross-domain problem with SMEs the student had to elicit	A real system at a partner organization the student did not choose
Authenticity	Chosen, familiar, safe to fail	Real problem, borrowed context	Real problem, real stakes, real client
What is new each time	The cycle itself	Eliciting the problem from others	Delivering under a partner's constraints

The instruction the cards carry: write the ladder into all three briefs, so a student sees LEN 6 and LEN 10 as harder runs of a cycle they already know, not as three separate assignments. Same machinery, rising stakes.

LEN 4: Evidence, Analytics, and Measurement

LEN 4 The shared LDT methods course *LEO-4 at practitioner depth (for LENS students) · taken across the program*

Owns	Decision-grade evidence for every LDT student. Ethical instrumentation and data governance, xAPI and Caliper, validity and reliability under constraint, designing the assessment before the content, capability over time and decay, appraising primary research, fairness beyond omission, gap attribution, judgment under inadequate evidence, communicating uncertainty. LENS builds it; LXD and AILE take it as their methods requirement. It is the literal measurement backbone of the program, and the place the analytics axis is shared across all three concentrations.
Not ours to redesign	This is the one card where LENS proposes rather than owns. LEN 4 serves three concentrations; LENS builds and stewards it but cannot bend it to LENS ends. What LENS does is inject: contribute the capability-measurement view (measurement of an interface, not only of a learner) as one strand among the course's general evidence training. The injection is additive and must not narrow the course to one concentration's domain. This is the correction from the earlier drafts, which wrongly treated LEN 4 as LENS-specific and overfull-for-LENS-to-fix.
Major topics	- What counts as evidence; designing the assessment before the content (taught fully here, not assumed from LEN 1, because most students never took LEN 1). - Ethical instrumentation and data governance; xAPI, Caliper, interoperability. - Validity, reliability, and capability over time and decay. - Gap attribution and fairness beyond omission. - Judgment under inadequate evidence; appraising primary research; communicating uncertainty. LENS injection: a capability-measurement thread, measuring whether a system builds the capability it promises, offered as one applied lens, not a required reframing of the whole course.
Assignments	A measurement design for a construct the student chooses, with instruments, validity argument, and limits. An evidence appraisal of a piece of primary research, judging what it does and does not support. A communicating-uncertainty brief for a non-technical audience.
PLO coverage	Assess: 6.1, 6.2, 6.3, 6.4 (all of data-informed design, this is PLO 6's home); 4.1, 4.3 (communication of evidence); 2.3 (equity, via fairness). Reinforce: 1.4 (justify on evidence).
LEO coverage	Assess (for LENS students): all of LEO-4 (4.1–4.8) at practitioner depth. This is where the attainment profile draws its LEO-4 evidence. Note: the LEO-4 join applies to LENS students only; LXD and AILE students take the same course but are assessed against program objectives, not LEOs. The record must state this so the profile does not claim non-LENS students.

AILE / LXD	Shared by definition, not offered outward, because it is already everyone's. This is the deepest convergence in the program: the one course all three concentrations take together. Its structural aperture (every LDT student passes through it) makes it the single largest lever for LENS visibility, since LENS builds it. That is an argument for building it exceptionally well, not for claiming it.
If we must cut	Not LENS's cut to make; the course serves three concentrations and its cut-line is a program decision. What LENS controls is the injection: if the capability-measurement thread crowds the general training, the thread is trimmed first, never the shared core. The LENS contribution is the most cuttable thing in the course, by design.
Flags	Doc 2 §5.3 wrongly labels LEN 4 LENS-specific; that is a record correction reaching the layout and aperture plans. The title "for High-Consequence Domains" must drop, because a course every LXD student takes cannot be named for one concentration's domains. Computational analytics moves to LEN 9 to keep this course to six modules. The 4.8 relief the LEN 1 redesign offers helps LENS students only.

LEN 6: Applied Problem Solving in Learning Engineering

LEN 6 The LENS-built methods course *LEO-1, 2, 5 · the second rung of the iteration ladder · prereq LEN 1*

Owns	Problem-solving method, on someone else's problem. Problem framing, stakeholder analysis, needs assessment, SME elicitation, and one full iteration on an authentic cross-domain problem the student did not invent. Where LEN 1 iterates on a system you know, LEN 6 makes you elicit the problem from experts first, then iterate. It also owns the delivery-and-support layer as a design skill and accessibility and inclusion. This is the LENS-built methods course, the counterpart to the shared LEN 4.
Major topics	● Problem framing and structured reasoning; the systems-thinking "frame" grain (LEN 1 sees, LEN 3 models, LEN 6 frames). ● Stakeholder analysis and needs assessment. ● SME elicitation: getting a capability problem out of people who hold it tacitly. ● Two cross-domain problems; one carried through a full iteration. ● The delivery-and-support layer (2.6) as a design skill on a bounded problem (the produced-artifact version is LEN 10). ● Accessibility and inclusion (2.7).
Assignments	● A problem-framing brief from a stakeholder elicitation the student conducts. ● One full iteration on an authentic cross-domain problem: framed, instrumented, evaluated, revised. ● A delivery-and-support design for the intervention, bounded.
PLO coverage	Assess: 1.1 (differentiate theories in a real problem); 2.1, 2.4 (learner needs, accessible design); 3.1, 3.5 (collaboration, evidence-based solutions). Reinforce: 6.1 (collect data), from LEN 4/LEN 1.
LEO coverage	Assess: 1.1 (capability requirements from elicitation); 2.1-2.7 (iterative development, including the delivery layer 2.6 and accessibility 2.7); 5.x (context and constraints). Reinforce: 1.2, 1.3 (system framing, from LEN 3).
AILE / LXD	LENS-centered: SME elicitation and cross-domain problem solving are the concentration's own methods. But the needs-assessment and stakeholder-analysis toolkit is genuinely useful to an LXD designer, and the course is a reasonable elective for a design-minded student who wants the eliciting-and-framing skills. Offerable outward selectively, not core to a sibling concentration.
If we must cut	Drop from two cross-domain problems to one, keeping the full iteration. The delivery-and-support layer compresses to a reading if LEN 10 is reliably carrying the produced-artifact version. What must not be cut: SME elicitation, which is the skill that distinguishes LEN 6 from LEN 1's iteration, and the full iteration itself, which is the ladder's second rung.
Flags	Prerequisite change: LEN 6 now assumes LEN 1's iteration cycle and measurement plan, so concurrent enrollment with LEN 1 no longer works; make LEN 1 a hard prerequisite (LEN 2 and 3 may stay concurrent). Bound the delivery-layer topic against LEN 10 so 2.6 is not taught twice at the same grain.

LEN 10: Learning Engineering Project

LEN 10 The capstone *all five LEOs · the third rung of the ladder · receives from LEN 8 (Move B)*

Owns	The end-to-end project at a partner organization , and the attainment profile. Boundary specification, unit-of-analysis commitment, three implementation weeks, one documented iteration, a sustainment plan, project management exercised, three signature deliverables (evidence dashboard, reproducible report, governance plan). It assesses LEO-3 rather than asserting it. It is the third and hardest run of the iteration ladder: a real system, a real client, constraints the student did not set.
Receives, not offloads	The capstone is not overfull and does not shed. It receives. Move B: implementation-science content from LEN 8, adoption, sustainment, why evidence does not move itself, arrives here as lectures alongside the project, so a student who never took LEN 8 as an elective still meets its load-bearing ideas at the moment they matter, standing up a real intervention. A capstone can lecture; that is what makes LEN 8 safe to be a true elective. This is the mechanism that keeps essentials out of skipped electives without overloading the core.
Major topics	● Problem identification at the partner; boundary specification (Wk 4); unit-of-analysis commitment (Wk 6). ● Three implementation weeks: the documented iteration, run for real. ● Received lectures (from LEN 8): adoption, sustainment, total cost of ownership, why evidence stalls, recommendations to leadership. ● Deliverables and the final briefing.
Assignments	● The three signature deliverables: evidence dashboard, reproducible report, governance plan. ● A sustainment plan (carrying the received LEN 8 material): how the intervention survives after the student leaves. ● The final briefing to the partner and a mixed audience.
PLO coverage	Assess : spans all six at capstone grain: 1.5 (deploy), 2.2 (systems), 3.1–3.5 (collaboration and leadership), 4.1–4.3 (communication), 5.3–5.4 (equitable systems), 6.2, 6.4 (data-informed).
LEO coverage	Assess : all five LEOs; this is where the attainment profile is aggregated and where LEO-3 (human-system collaboration) is assessed for real rather than asserted. Consequence : the profile aggregates from every course's CLO-to-LEO join, and those joins changed in LEN 1 and will change in LEN 3 and LEN 5. The profile must be rebuilt after each course is designed, not once at the end.
AILE / LXD	None. The capstone assesses the LENS attainment profile and cannot be offered outward. It is the course that certifies the degree.
If we must cut	A capstone is bounded by the project, not the syllabus, so there is little to cut. If the received LEN 8 lectures crowd the project, they compress to two sessions plus a reading, but they do not leave, because they are the only place a non-LEN-8 student meets sustainment. Project management is exercised here and taught nowhere else, so it stays.
Flags	Depends on a partner pipeline (APL practicum, sponsored problems); that pipeline is a real dependency and a recruiting asset. The three-iteration ladder should be stated in the capstone brief. If the concentration wants a graded failure-analysis artifact in the portfolio and LEN 3's is not enough, the governance plan here can carry one.

What Part 3 commits, and what Part 4 does

- **LEN 4 is a contribution, not a possession.** LENS builds and stewards the shared methods course and injects a capability-measurement thread; it does not redesign it and its cut-line is a program decision. The record correction (doc 2 §5.3) and the title change are named.
- **The iteration ladder is now explicit across LEN 1, 6, 10.** Same cycle, escalating authenticity. Write it into all three briefs.
- **Move B is concrete:** LEN 10 receives LEN 8's essentials as lectures, which is what lets LEN 8 be a true elective. A capstone that lectures is the mechanism.
- **The attainment profile is a moving target.** It rebuilds after each course is designed, because every CLO-to-LEO join feeds it and the joins are changing.

Part 4 takes the four electives (LEN 5, 7, 8, 9), where the depth lives and where the AILE cross-listing question (LEN 9) and the folded-essentials sources (LEN 7, LEN 8) are settled. Part 5 consolidates every cut-line into one table and lists the record changes across docs 1 and 2.

PART 4

The Four Electives: LEN 5, 7, 8, 9

The elective problem, and the rule that solves it

Students take two electives. If LEN 5 and LEN 9 are the recommended pair, most students never take LEN 7 or LEN 8. That is fine only if nothing essential lives solely in 7 or 8. Parts 1–3 enforced exactly that: LEN 7's governance and bias folded into LEN 2 and LEN 3; LEN 8's implementation essentials arrive in LEN 10 as lectures. So these four cards describe depth courses, and each names what it already gave away.

Elective	Standing	Gave forward	What remains here (the depth)
LEN 5	Recommended	Failure methods to LEN 3 at first-contact grain	Capability analysis and failure analysis at method depth
LEN 7	Elective	Governance to LEN 2; bias/fairness to LEN 3	The full governance, risk, and equity-impact treatment
LEN 8	Elective	Implementation essentials to LEN 10 as lectures	The full implementation-science and organizational-learning treatment
LEN 9	Highly recommended	Nothing (it receives LEN 4's computational analytics)	Computational and AI methods at practitioner depth

The tension you named: LEN 5 and LEN 9 both matter enough to recommend, and that is two slots, leaving no room for 7 or 8. The resolution is not to demote 5 or 9; it is that 7 and 8 are safe to be optional because their essentials are already in the required courses. A student who takes 5 and 9 has met governance, bias, and implementation in the core. A student who wants depth in those takes 7 or 8 instead of 5 or 9, a real choice, not a gap.

LEN 5: Human Capability and Failure Analysis

LEN 5 Elective, recommended *LEO-1, 5 · retitled from Human Capability Analysis and Requirements · builds on LEN 6*

Owns	The deep analytical toolkit, forward and backward. Two halves that are one stance: analyzing what a capability requires (cognitive task analysis, critical decision method, requirements elicitation) and analyzing how a capability failed (the established failure-analysis methods at depth). LEN 3 codes a failure against the casebook taxonomy; LEN 5 teaches the methods behind the codes. The retitle reflects the second half, added when LEN 1 shed failure methods.
Major topics	● Capability analysis: cognitive task analysis, critical decision method, requirements elicitation in operational settings. ● Extending elicitation into unfamiliar domains; translating findings into measurable requirements. ● Failure analysis (from the LEN 1 shed): Reason (human error), Hollnagel (Safety-I/Safety-II), Leveson (STAMP), at method depth. ● Method selection and justification: choosing the right analytic method for a case and defending the choice. ● The analysis as a produced artifact.
Assignments	● A cognitive task analysis of a real operator role. ● A failure analysis of a documented case using a named method, with the method choice justified. ● A requirements translation: from the analyses to measurable capability requirements.
PLO coverage	Assess: 1.3 (analyze/evaluate design options); 6.3 (mixed-methods analysis); 5.1 (analyze how systems influence). Reinforce: 1.1 (differentiate theories), from LEN 6.
LEO coverage	Assess: 1.1, 1.4 (capability requirements and analysis at depth); 5.x (constraints, via failure analysis). This is LEO-1's method-depth home, complementing LEN 3's modeling-depth home. Between them they carry LEO-1 to full practitioner depth.
AILE / LXD	LENS-only. Cognitive task analysis and failure methods are the concentration's own analytic core. Not offerable

	outward; this is a course that certifies the specialization's method.
If we must cut	As an elective it is cut by not being taken, not by shedding topics. If the failure-analysis artifact is judged essential to the portfolio, LEN 5 moves from recommended toward required, which is the decision below. Internally, method-selection compresses before either half.
Flags	The real decision: LEN 5 is described as recommended, but it is the only method-depth home for both capability analysis and failure analysis. If either is essential to what a LENS graduate must be able to do, LEN 5 is closer to required than recommended. Decide this explicitly rather than letting it be settled by enrollment. Retitle touches the catalogue.

LEN 7: Bias, Risk, and Governance in Learning System Design

LEN 7 Elective *LEO-4, 5 · depth home for what folded into LEN 2 and LEN 3 · recommended LEN 4 first*

Owns	The full governance, risk, and equity treatment. Algorithmic bias and fairness definitions, surveillance and institutional accountability, governance structures in regulated environments, cross-regime and platform-dependency governance, risk assessment, equity-impact evaluation, the governance plan. The essentials of this now live in LEN 2 (governance) and LEN 3 (bias/fairness); LEN 7 is where a student who wants the whole discipline gets it.
Already given forward	Move A took the essentials into the core. Governance and accountability, and the governance-objection diagnostic, are required topics in LEN 2. Bias, fairness, and measurement equity are required in LEN 3. So LEN 7 is safe to be an elective most students skip: no graduate misses governance or fairness by skipping it. What LEN 7 adds is depth, breadth of regime, and the full governance-plan artifact, not the existence of the ideas.
Major topics	● Algorithmic bias and the competing fairness definitions (deeper than LEN 3's measurement-equity treatment). ● Surveillance, accountability, and institutional responsibility. ● Governance structures in regulated environments; cross-regime and platform-dependency governance. ● Risk assessment for deployments; the governance-objection diagnostic at depth (introduced in LEN 2). ● Equity-impact evaluation and the full governance plan.
Assignments	● A governance plan for a real or proposed learning system in a regulated environment. ● An equity-impact evaluation with a mitigation design. ● A risk assessment distinguishing dissolvable objections from do-not-deploy ones.
PLO coverage	Assess: 5.2, 5.3, 5.4 (power dynamics, mitigation, equitable systems); 2.3 (equity evaluation). Reinforce: 6.4 (data collection for evaluation), from LEN 4.
LEO coverage	Assess: 5.x (navigating sociotechnical constraints, at full depth); 4.x fairness subobjectives. Note the LEO 1.5 question: the governance-objection diagnostic is currently LEO 1.5 (a Systems Analysis subobjective) but LEN 7 maps LEO-4 and 5. Either LEN 7 gains LEO-1, or 1.5 re-homes under LEO-5 in doc 1, where a governance skill arguably belongs. A doc 1 decision, flagged in Part 5.
AILE / LXD	Converges strongly with AILE, and the leading cross-list candidate after LEN 2. AILE leaders govern exactly these systems; LEN 7 teaches governance as design, which AILE does not. Offerable to AILE, and a natural place for AILE-LEN co-enrollment. For LXD, the equity material overlaps F2 (F2 critiques, LEN 7 designs).
If we must cut	Cut by not being taken; its essentials are already safe in the core. If offered and overfull, cross-regime governance compresses before bias or the governance plan. The governance plan artifact is the course's signature and stays.
Flags	The LEO 1.5 re-homing (doc 1). The F2 division must be stated so LEN 7 and F2 do not read as one course taught twice. Strong AILE cross-list candidate; that conversation is Diamond's and the AILE lead's.

LEN 8: Knowledge Transfer and Organizational Learning

LEN 8 Elective *LEO-2, 5 · depth home for what LEN 10 receives as lectures*

Owns	The implementation-science course: why evidence does not move itself. Knowledge transfer mechanisms, organizational learning theory, implementation frameworks (EPIS, CFIR, Active Implementation), adoption and sustainment, total cost of ownership, recommendations to leadership. It answers the question every practitioner eventually hits: I have the evidence, why will nothing change? Its essentials arrive in LEN 10 as lectures; LEN 8 is the
------	--

	full discipline.
Already given forward	Move B took the essentials into the capstone. Adoption, sustainment, and why evidence stalls arrive in LEN 10 as received lectures, at the moment a student is standing up a real intervention. So a graduate who never takes LEN 8 still meets its load-bearing ideas. LEN 8 adds the frameworks (EPIS, CFIR, Active Implementation), the organizational-learning theory, and the depth, not the existence of the ideas.
Major topics	● Knowledge transfer mechanisms; tacit-to-explicit, expert-to-novice. ● Organizational learning theory. ● Implementation-science frameworks: EPIS, CFIR, Active Implementation. ● Adoption, sustainment, and total cost of ownership. ● Recommendations to leadership; the research-practice gap and how to close it.
Assignments	● An implementation plan for a real intervention using a named framework. ● A sustainment and total-cost analysis. ● A recommendation memo to leadership making the case for adoption on evidence.
PLO coverage	Assess: 3.4, 3.5 (evaluation leadership, evidence-based solutions); 4.2, 4.3 (justify to audiences, communicate with parties). Reinforce: 6.2 (revise on data).
LEO coverage	Assess: 2.x (iterative development, sustainment side); 5.x (constraints of real deployment).
AILE / LXD	The most under-recognized outward course, and the strongest LXD recommendation. It is what an LXD designer needs and rarely gets: why a good design does not survive deployment. It is also directly in the new dean's vocabulary (implementation science, fidelity, sustainment). Offerable to both siblings; recommend it to LXD students taking LENS electives.
If we must cut	Cut by not being taken; essentials safe in LEN 10. If offered and overfull, total-cost-of-ownership compresses before the frameworks. The frameworks are the course's spine and stay.
Flags	Reposition in advising: doc 2 recommends LEN 5 and LEN 9 to LENS students, which is right, but LEN 8 is the better recommendation for a cross-concentration LXD or AILE student. Name that in the advising section (Part 5).

LEN 9: Computational and AI Methods for Learning Engineers

LEN 9 Elective, highly recommended *LEO-4, 3 · receives LEN 4's computational analytics · the Whiting bridge*

Owns	Computational and AI methods at practitioner depth, and the AI-practitioner competency. NLP for learning data, ML for learner modeling, LLM-as-judge evaluation designs, computational analytics, building tools as creative partners, assessing computational impact on evidence quality. This is where LENS's AI-practitioner stance (Part 1) gets its computational teeth: a graduate who can instrument an AI tutor, judge an LLM-as-evaluator design, and say whether an AI intervention built capability or eroded judgment.
Why highly recommended	Two reasons it sits above the other electives. First, the job market: computational and AI-methods skills are where learning-engineering hiring is growing, and this is the course that produces them. Second, it is the Whiting bridge, the concrete form of the engineering-school partnership LENS needs, and the home of the AI-practitioner competency that keeps LENS from ceding AI to AILE. Highly recommended is a market judgment as much as a curricular one.
Major topics	● NLP for learning data; ML for learner modeling. ● LLM-as-judge evaluation designs; when an AI evaluator is trustworthy and when it is not. ● Computational analytics (received from LEN 4): learning-analytics pipelines beyond the basics, computational evidence synthesis. ● Building tools as creative partners. ● Assessing computational impact on evidence quality.
Assignments	● A learner-modeling or NLP analysis on real learning data. ● An LLM-as-judge evaluation design with a validity argument for the AI evaluator. ● A tool build with an assessment of its impact on evidence quality.

PLO coverage	Assess: 6.1, 6.3, 6.4 (data collection, analysis, comprehensive data efforts); 1.2 (use digital technologies to build). Reinforce: 3.3 (quality of technology-supported experiences).
LEO coverage	Assess: 4.x (data and measurement, computational side); 3.x (human-system collaboration, where AI-as-partner gets computational depth).
AILE / LXD	Converges with AILE and is the Move 2 cross-list candidate. AILE has the Whiting partnership; LENS has the computational elective; a Whiting data-science course already substitutes. Cross-listing LEN 9 with AILE converts the partnership from aspiration to a course. That scope is Diamond's and the AILE lead's to set.
If we must cut	Cut by not being taken, though "highly recommended" is meant to make that rare. If a Whiting data-science course substitutes, the LEN 4 computational transfer has no LENS home for that student, a small gap to note. Internally, tool-building compresses before the analytics core.
Flags	Highly-recommended standing plus LEN 5 recommended is effectively both elective slots; confirm that is intended (Part 5 revisits the two-slot math). The AILE cross-list (Move 2) is the live external decision. Receives LEN 4's computational analytics.

What Part 4 settles, and what Part 5 does

- **The elective rule held:** every essential is in a required course, so 7 and 8 are safe to skip. Each card names what it gave forward.
- **LEN 5 is closer to required than recommended**, because it is the only method-depth home for capability and failure analysis. Decide it, do not let enrollment decide it.
- **LEN 9 is highly recommended on job-market grounds**, and is the AI-practitioner competency and the Whiting bridge.
- **The two-slot math is real:** 5 recommended plus 9 highly recommended is both slots. That is defensible if 7 and 8 are genuinely depth-optional, which the fold-forwards make true. Part 5 states it plainly so it is a choice, not an accident.
- **Two external decisions surfaced:** LEN 7 and LEN 9 as AILE cross-lists, both Diamond-and-AILE-lead calls, not LENS's to execute.

Part 5 consolidates all ten cut-lines into one table, lists the record changes across doc 1 (LEO 1.5 re-home; LEO-4/LEN 1) and doc 2 (\$5.3 shared-methods correction; LEN 4 title; the fold-forwards; the mappings), settles the two-elective math and the LEN 5 required-vs-recommended question, and gives the advising table for cross-concentration students.

Consolidation and Decisions

1. The cut-lines, all ten courses

Each course names what leaves first under load and what must not leave, so the concentration can be reduced deliberately rather than by attrition. “Protected” is the floor: cutting it changes what the course is.

Course	Cuts first	Protected (must not cut)
LEN 1	M3 compresses to one week; then M2 sociocultural to a reading	M4 measurement (the spine); the M5 iteration cycle (the build)
LEN 2	Facilitators-as-roles (3.7); then a teaming subtopic	The folded governance material
LEN 3	Failure-methods first contact to one example; then a domain studio (3 to 2)	The folded bias/fairness material; the system-modeling core
LEN 4	Not LENS’s cut; the injected capability thread trims before the shared core	Nothing LENS-owned; it is a shared course
LEN 5	Method-selection compresses; cut mainly by not being taken	Both halves: capability analysis and failure analysis
LEN 6	Two cross-domain problems to one; delivery layer to a reading	SME elicitation; the full iteration
LEN 7	Cross-regime governance compresses; cut mainly by not being taken	The governance-plan artifact
LEN 8	Total-cost-of-ownership compresses; cut mainly by not being taken	The implementation frameworks
LEN 9	Tool-building compresses before the analytics core	The computational analytics from LEN 4; the AI-practitioner competency
LEN 10	Received LEN 8 lectures to two sessions plus a reading	The project, the deliverables, project management

In every required course the protected floor is the course’s reason to exist or a folded-in essential. In every elective the essentials are already elsewhere, so an elective is cut by not being chosen, not by shedding topics.

2. Record changes

Doc 1 (Five Competencies / LEOs)

- **LEO 1.5 re-home.** The governance-objection diagnostic is LEO 1.5 (Systems Analysis) but its home is LEN 7 (LEO-4, 5). Re-home 1.5 under LEO-5, or add LEO-1 to LEN 7’s mapping; re-homing is cleaner.
- **LEO-4 introduced in LEN 1.** The redesign opens LEO 4.8, 4.1, 4.5 at awareness in LEN 1. The attainment-profile join (decisions/02) records LEN 1 as an introduce-depth carrier of LEO-4.

Doc 2 (Objectives and Course Mapping)

- **§5.3 correction.** Record LEN 4 as the shared LDT methods course and LEN 6 as the LENS-built one.
- **LEN 4 title.** Drop “for High-Consequence Domains”; content unchanged.
- **Move A.** LEN 2 gains governance and accountability; LEN 3 gains bias, fairness, and measurement equity; LEN 7 remains the depth home for both.
- **Move B.** LEN 10 receives LEN 8 implementation content as lectures; note it in the LEN 10 description.
- **Transfers.** Computational analytics from LEN 4 to LEN 9; failure-analysis methods from LEN 1 to LEN 5 (depth) and LEN 3 (first contact).
- **LEN 5 retitle.** “Human Capability and Failure Analysis.”
- **Mapping updates and version.** LEN 1 five CLOs with PLO-6 and LEO-4 added; LEN 2 LEO-3 plus LEO-5; LEN 3 LEO-1 plus LEO-5; LEN 5, 7, 8, 9 per their cards. Bump doc 2 to v2.8 with a changelog entry.

3. Elective structure and the two-slot decision

Students take two electives from LEN 5, 7, 8, 9, or a university substitute. The recommended pair is LEN 5 and LEN 9. That pair consumes both slots, so the plan depends on LEN 7 and LEN 8 being genuinely optional, which the fold-forwards make true.

Elective	Standing	Basis
LEN 9	Highly recommended	Job market (computational and AI-methods hiring) and the Whiting bridge; the AI-practitioner competency.
LEN 5	Recommended; decide if required	The only depth home for capability and failure analysis. If either is essential to a LENS graduate, LEN 5 is required.
LEN 7	Elective	Depth in governance, risk, equity; essentials in LEN 2 and LEN 3. AILE cross-list candidate.
LEN 8	Elective	Depth in implementation science; essentials in LEN 10. Strongest cross-concentration LXD recommendation.

The two-slot math: LEN 5 plus LEN 9 is both slots, and a student taking them has met governance, bias, and implementation in required courses. A student wanting depth in those takes LEN 7 or LEN 8 in place of one. Both are complete paths.

4. Decisions required

#	Decision	Owner
1	Does LEN 1 launch on the redesigned five-CLO syllabus?	Program director
2	Is LEN 5 required rather than recommended?	Program director
3	Confirm Move A: governance into LEN 2, bias/fairness into LEN 3.	Program director
4	Confirm doc 2 corrections: \$5.3 labeling; LEN 4 title.	Program director
5	LEO 1.5 re-home under LEO-5 (doc 1).	Program director
6	Cross-list LEN 9, and possibly LEN 7, with AILE.	Program director and AILE lead

Decisions 1 and 2 gate the build and the elective structure; the rest confirm moves specified on the course cards.

5. Cross-concentration advising

For LXD and AILE students taking LENS courses as electives; extends doc 2 §6.5, which advises LENS students only.

Student	Recommended LENS pair	Rationale
LXD	LEN 8 + LEN 2	Why evidence does not move into practice (LEN 8); interface and alert design (LEN 2). LEN 4 is already their methods requirement.
AILE, building	LEN 2 + LEN 7	Design the human-machine collaboration, then govern it.
AILE, technical	LEN 2 + LEN 9	The collaboration, then the computational methods behind it.
Either, one course	LEN 1 (full or primer)	The frame; everything else in LENS assumes it.

6. Build sequence, once decisions are made

- **LEN 1 M3-M6**, measurement module first. Modules 1 and 2 are built.
- **LEN 2**, carries the folded governance material.
- **LEN 3 and LEN 5 together**, the failure-methods split binds them; LEN 3 carries the folded bias material.
- **LEN 4 injection**, coordinated with the shared-course owner.
- **LEN 6, 10, then electives**. The attainment profile rebuilds after each course.

Each course receives a design pass before a build pass.